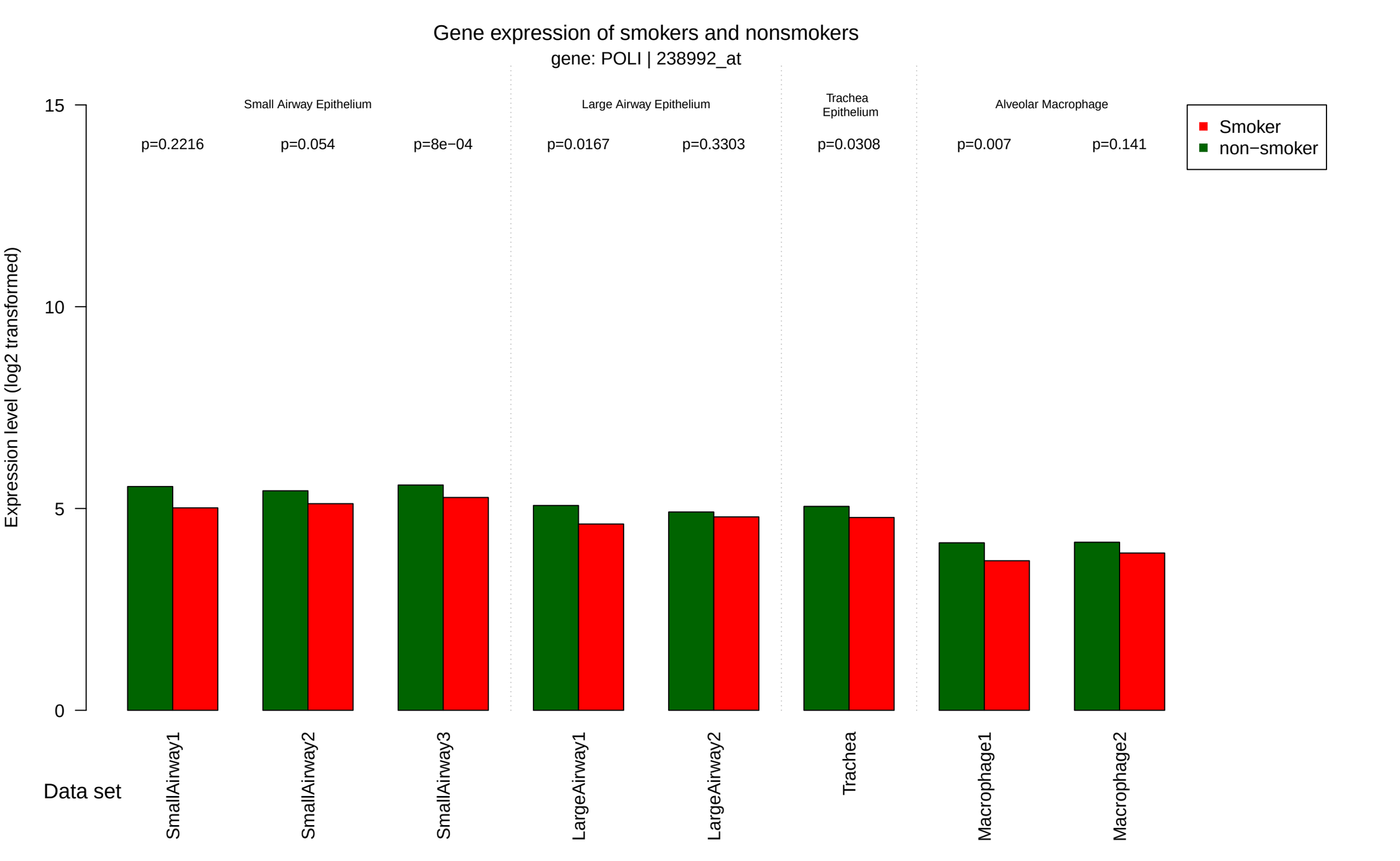
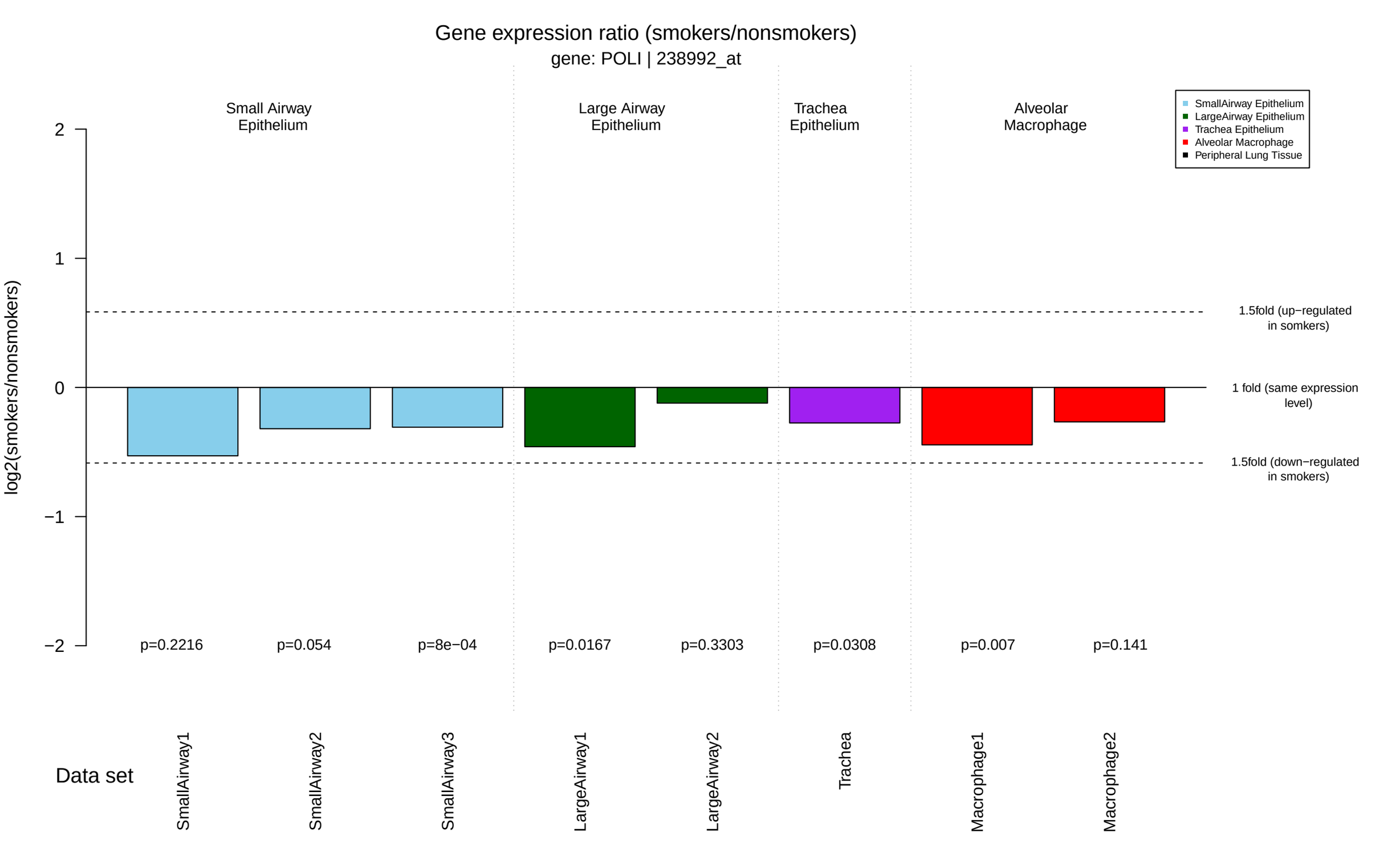
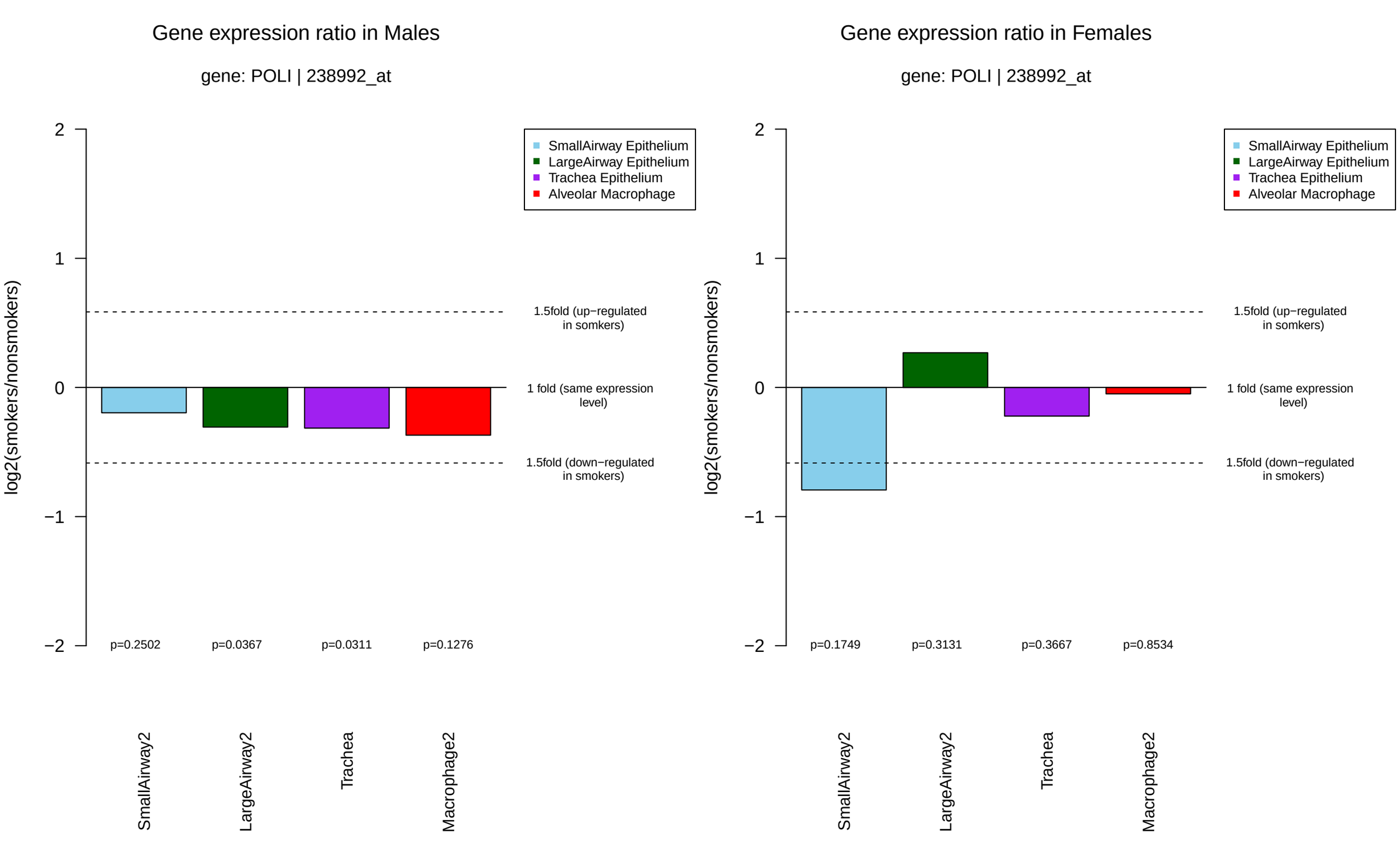
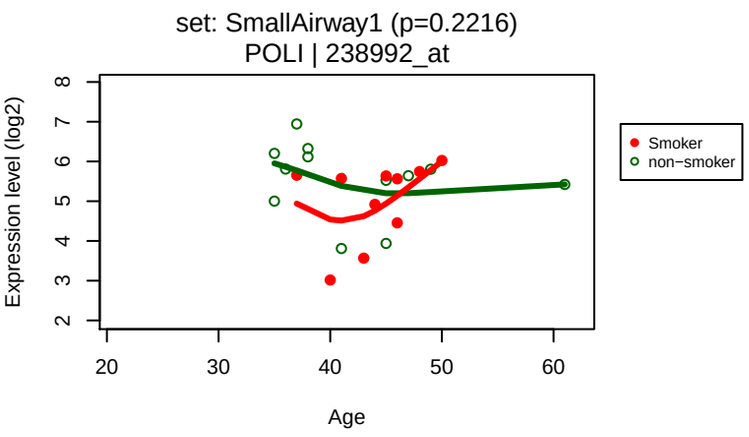
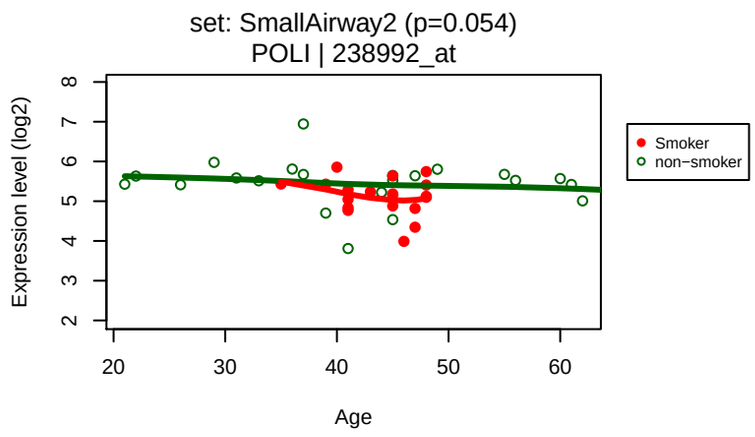
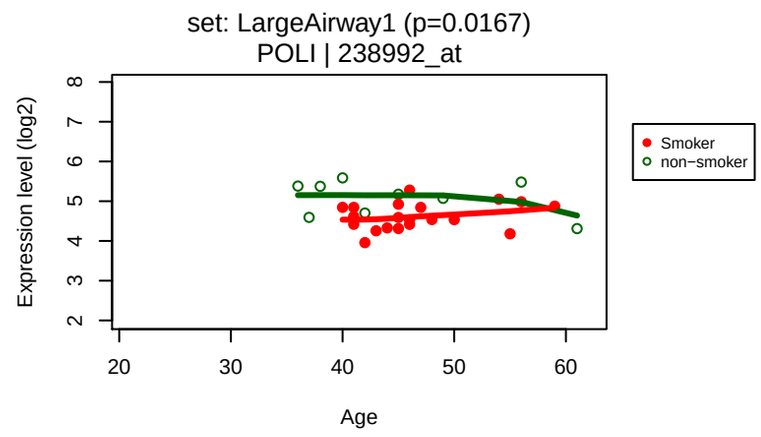

Gene expression of smokers and nonsmokers
gene: POLI | 238992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2216
p=0.054
p=8e−04
p=0.0167
p=0.3303
p=0.0308
p=0.007
p=0.141
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

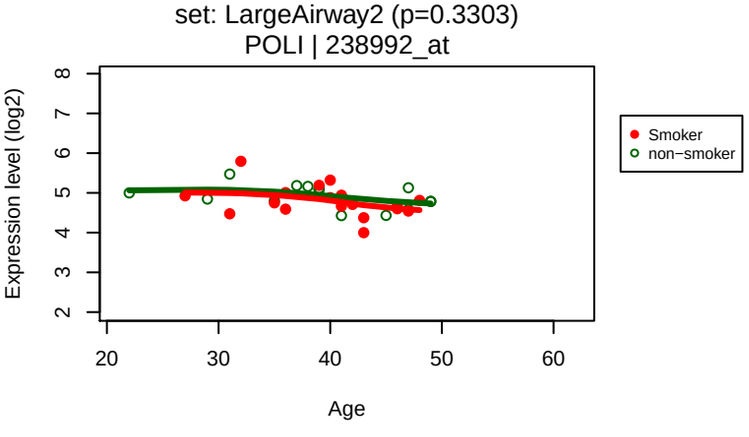

Gene expression ratio (smokers/nonsmokers)
gene: POLI | 238992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2216
p=0.054
p=8e−04
p=0.0167
p=0.3303
p=0.0308
p=0.007
p=0.141
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

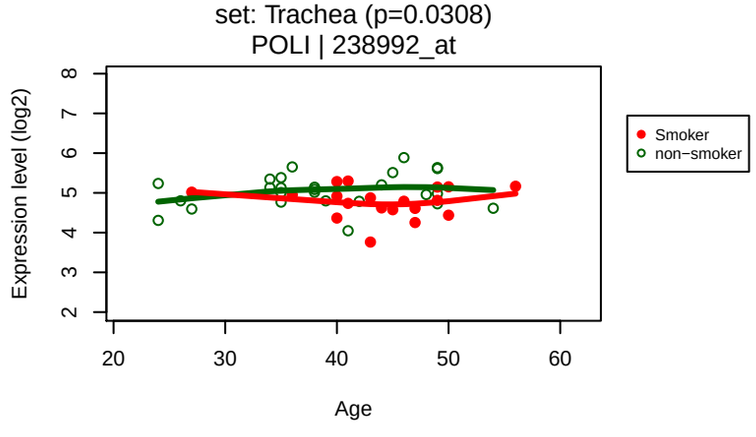

Gene expression ratio in Males
gene: POLI | 238992_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2502
p=0.0367
p=0.0311
p=0.1276
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLI | 238992_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1749
p=0.3131
p=0.3667
p=0.8534
SmallAirway2
LargeAirway2
Trachea
Macrophage2

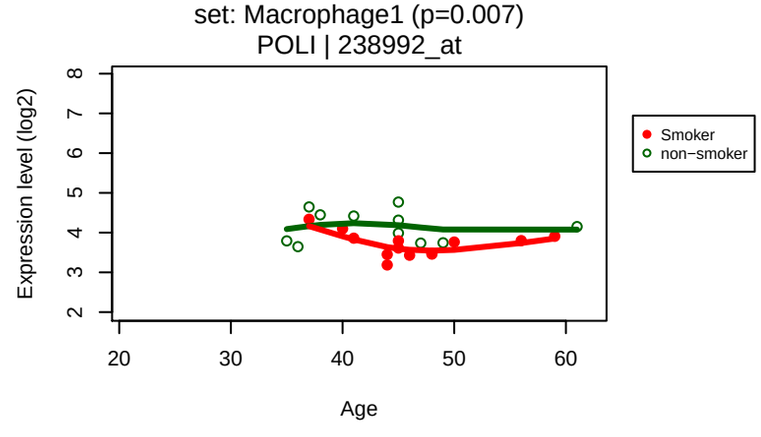

set: SmallAirway1 (p=0.2216)
POLI | 238992_at
Expression level (log2)
Age
Smoker
non−smoker

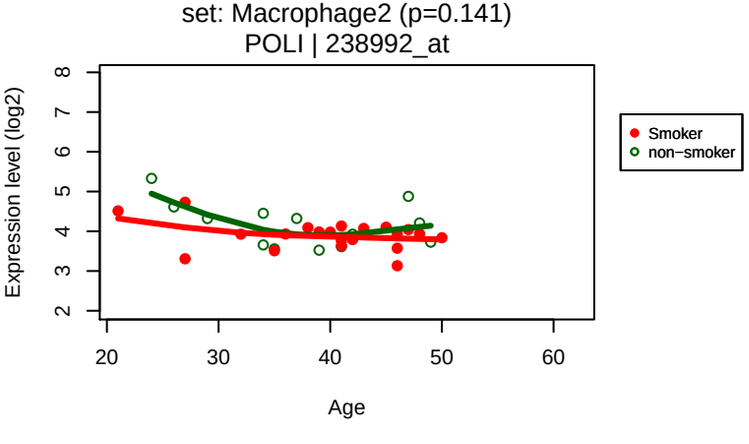

set: SmallAirway2 (p=0.054)
POLI | 238992_at
Expression level (log2)
Age
Smoker
non−smoker

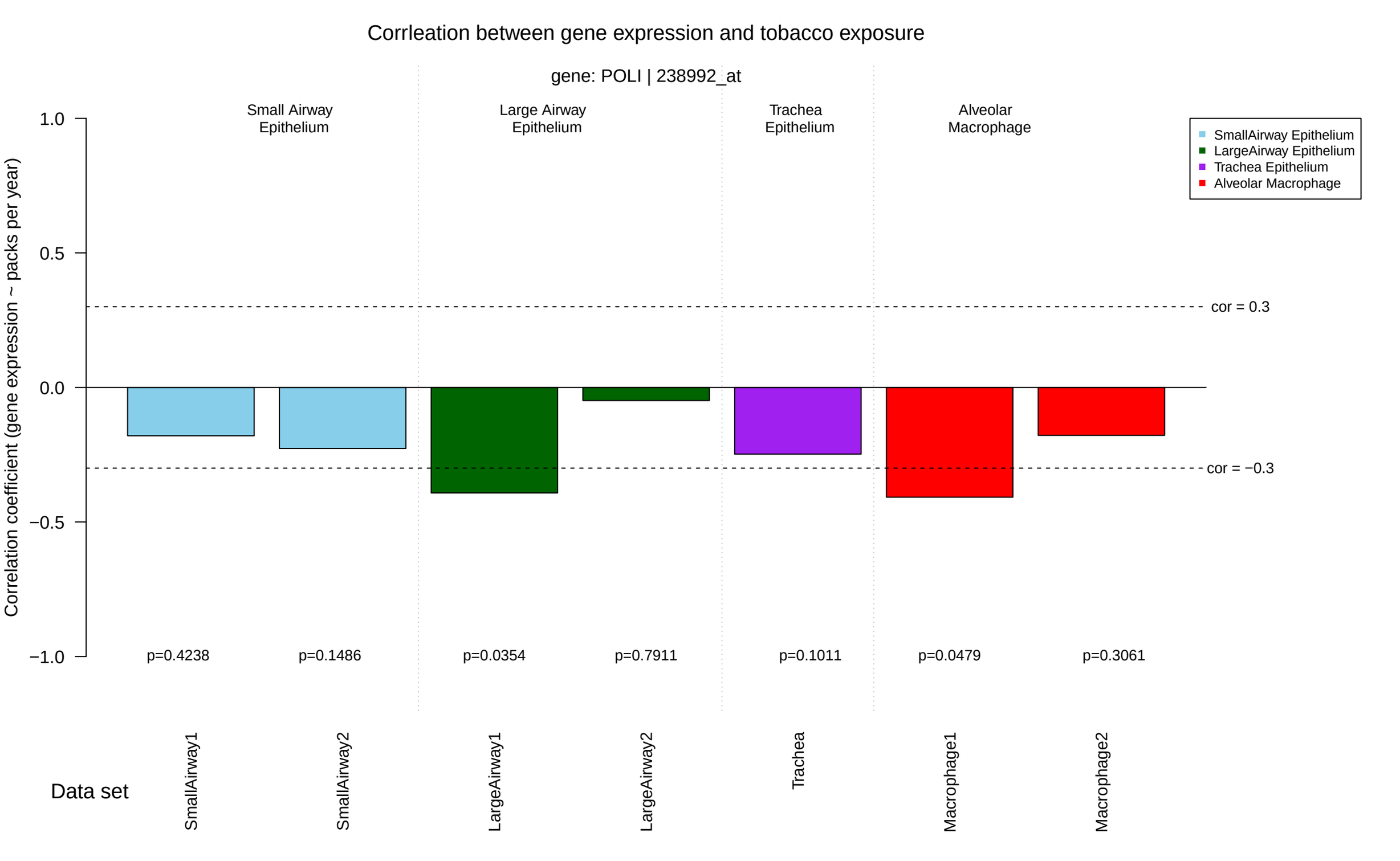

set: LargeAirway1 (p=0.0167)
POLI | 238992_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3303)
POLI | 238992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0308)
POLI | 238992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.007)
POLI | 238992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.141)
POLI | 238992_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLI | 238992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4238
p=0.1486
p=0.0354
p=0.7911
p=0.1011
p=0.0479
p=0.3061
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2